

Mathématiques : CM

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1 Sequences of functions

Let I be an interval of $\mathbb{R}$ (f_n) a sequence of functions on I .

Sequence whose general term f_n is a function $f_n : I \rightarrow \mathbb{R}$.

Modes of convergence of (f_n) to f where $f : I \rightarrow \mathbb{R}$.

Pointwise convergence :

(f_n) converges pointwise to a function f if $\forall x \in I, f_n(x) \xrightarrow{n \rightarrow +\infty} f(x)$

Example : on $I = [0, 1], f_n(x) = x^n$

$$\text{Let } x \in [0, 1], \begin{cases} \text{if } x < 1, x^n \rightarrow 0 \\ \text{if } x = 1, x^n \rightarrow 1 \end{cases}$$

Thus $f_n(x) \xrightarrow{n \rightarrow +\infty} f(x)$ with :

$$f(x) = \begin{cases} 0 & \text{if } x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

$(f_n) \xrightarrow{\text{p.w}} f$ (i.e converges pointwise)

2 Uniform convergence

Supremum of a bounded set S .

Let S be a set, upperbounded.

$$\exists M \in \mathbb{R}, \forall x \in S, x \leq M$$

Supremum of S : the smallest possible M denoted $\text{Sup}S$.

Example : $S = [0, 3], \text{Sup}S = 3$

$S_2 = [0, 3[, \text{Sup}S = 3$

Remark : For $[0, 3], \text{Sup}[0, 3] = 3 = \max[0, 3]$

For $[0, 3[= \text{no maximum value } (3 \notin [0, 3[) \text{ but } \text{Sup}[0, 3] = 3$

3 Definition : norm $\|\cdot\|_\infty$

Let E be the vector space of bounded function on I .

Let $f \in E$,
TODO

3.1 Properties

$$\forall (f, g) \in E^2, \forall \alpha \in \mathbb{R}$$

$$\|f\|_\infty = 0 \implies f = 0_E$$

$$\|\alpha f\|_\infty = \|\alpha\| \times \|f\|_\infty$$

$$\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty$$

(f_n) converges uniformly to f if $\forall n \geq n_0, f_n - f$ is bounded $\implies \|f_n - f\|_\infty \rightarrow 0$

Example : on $[0, 1]$, $f_n(x) = x^n$.

We have seen that : $\forall x \in [0, 1], f_n(x) \rightarrow f(x) = \begin{cases} x^n & \text{if } x < 1 \\ 1 & \text{if } x = 1 \end{cases}$

$(f_n) \xrightarrow{\text{p.w}} f$ (i.e converges pointwise)

Does (f_n) converges uniformly to f ?

Let $n \in \mathbb{N}$. Let us study $f_n - f$:

$$\forall x \in [0, 1], f_n(x) - f(x) = \begin{cases} x^n & \text{if } x < 1 \\ 0 & \text{if } x = 1 \end{cases}$$

$$f_n(1) - f(1) = 1^n - 1 = 0$$

Thus (f_n) converges uniformly.

Example 2 : on $\mathbb{R}$, $f_n(x) = \frac{\sin(nx)}{n}$

Pointwise : let $x \in \mathbb{R}$

$$\forall n \in \mathbb{N}, -\frac{1}{n} \leq \frac{\sin(nx)}{n} \leq \frac{1}{n}$$

$$-\frac{1}{n} \rightarrow 0 \wedge \frac{1}{n} \rightarrow 0$$

Thus, $f_n(x) \rightarrow 0$

$f_n(x) \xrightarrow{p.w} f$ (zero function)

$\forall x \in \mathbb{R}, f(x) = 0$

Does (f_n) converges to f ?

Let $n \in \mathbb{N}^*$, study $f_n - f$

$$\forall x \in \mathbb{R}, |f_n(x) - f(x)| = \left| \frac{\sin(nx)}{n} \right|$$

$$\sup |f_n(x) - f(x)|, x \in \mathbb{R} = \frac{1}{n}$$

$$\frac{1}{n} \rightarrow 0 \implies \|f_n - f\|_\infty \rightarrow 0 \implies (f) \xrightarrow{\text{CVG}} f$$

Thus we have a **Uniform convergence**.

4 L2 convergence (on a closed interval $[a, b]$).

4.1 Definition - piecewise continuity

A function $f = [a, b] \rightarrow \mathbb{R}$ is "piecewise" continuous on $[a, b]$ if it is continuous at all $x \in [a, b]$, except maybe at a finite set of values $\{d_1, d_2, \dots, d_n\}$

At each discontinuous point d , there is a left limit and a right limit.

$$f(d^-) = \lim_{\substack{x \rightarrow d \\ x < d}} f(x)$$

$$f(d^+) = \lim_{\substack{x \rightarrow d \\ x > d}} f(x)$$

Remark : we can define $\int_0^2 f(x) dx = \int_0^1 f(x) dx + \int_1^2 f(x) dx$

4.2 Norm $\|\cdot\|_2$ on $[a, b]$

Let E be the vector space of functions, f piecewise continuous on $[a, b]$

$$\|f\|_2 = \sqrt{\int_a^b f^2(x) dx}$$

Remark : $\|\cdot\|_2$ comes from the scalar product $\langle f, g \rangle = \int_a^b f(x)g(x) dx$

$$\|f\|_2 = \sqrt{\langle f, f \rangle}$$

4.3 Properties

$\forall (f, g) \in E^2, \forall \alpha \in \mathbb{R} :$

- $\|f\|_2 = 0 \implies f$ is the zero function (except maybe at a finite set of points)
- $\|\alpha f\|_2 = |\alpha| * \|f\|_2$
- $\|f + g\|_2 \leq \|f\|_2 + \|g\|_2$

4.4 Definition

(f_n) converges L2 to f if : $\|f_n - f\|_2 \xrightarrow{n \rightarrow +\infty} 0$

Example 1 :

On $[0, 1], f_n(x) = x^n :$

$$(f_n) \xrightarrow{\text{p.w}} f \text{ where } f(x) = \begin{cases} 0 & \text{if } x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

(f_n) does not converge to f

Does (f_n) converge L2 to f

Let $n \in \mathbb{N}$,

$$f_n(x) - f(x) = \begin{cases} x^n & \text{if } x < 1 \\ 0 & \text{if } x = 1 \end{cases}$$

5 Series of functions

5.1 Definition

Let (f_n) be a sequence of functions.

The series $\sum f_n$ is the sequence (S_n) :

$$\forall n \in \mathbb{N}, S_n = f_0 + f_1 + \dots + f_n = \sum_{k=0}^n f_k$$

Example : Power series $f_n : x \mapsto a_n x^n$, ($a_n \in \mathbb{R}$)

5.2 Definition

- $\sum f_n$ converges point wise if (S_n) converges point wise
- $\sum f_n$ converges uniformly if (S_n) converges uniformly
- $\sum f_n$ converges L2 if (S_n) converges L2

Necessary condition :

- $\sum f_n$ converges uniformly $\implies (f_n) \rightarrow f_n(x) = 0$ (converges uniformly)
- f_n does not converge to the zero function $\implies \sum f_n$ does not converge
- $\sum f_n$ converges L2 $\implies (f_n) \rightarrow f_n(x) = 0$ (converges L2)

Sufficient condition :

- $\sum \|f_n\|_\infty$ converge $\implies \sum f_n$ converge
- $\sum \|f_n\|_2$ converge $\implies \sum f_n$ converge L2

6 Fourier Series

Let f be piecewise continuous on $\mathbb{R}$ of period T .

$$\forall x \in \mathbb{R}, f(x + T) = f(x)$$

6.1 Case $T = 2\pi$

6.1.1 Definition : Fourier series of f

$$FS(f) = \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos(nx) + b_n \sin(nx))$$

$$\forall n \in \mathbb{N}, a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

6.1.2 Explanation

Let $\mathbb{E}$ the $\mathbb{R}$ vector space of piecewise continuous, functions, period.

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) g(x) dx$$

Note that :

$$\|f\|^2 = \langle f, f \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx$$

($\|f\|$ small means f^2 and $|f|$ is small on average)

Then :

$$\begin{cases} a_n = \langle f, \cos(nx) \rangle \\ b_n = \langle f, \sin(nx) \rangle \\ a_0 = \langle f, 1 \rangle \end{cases}$$

Let :

$$\mathcal{F}_n = \left(\frac{1}{\sqrt{2}}, \cos(x), \sin(x), \cos(2x), \sin(2x), \dots, \cos(nx), \sin(nx) \right)$$

$\mathcal{F}_n$ is an orthonormal family.

The partial sum S_n of the series is the orthogonal projection of f on $\text{Span}(\mathcal{F}_n)$ defined such as:

$$S_n = \frac{a_0}{2} + (a_1 \cos(x) + b_1 \sin(x)) + \dots + (a_n \cos(nx) + b_n \sin(nx))$$

S_n is the element of $\text{Span}(\mathcal{F}_n)$ which minimizes $\|f - S_n\|^2$

S_n is the function of the form :

$$\frac{a_0}{2} + (a_1 \cos(x) + b_1 \sin(x)) + \dots + (a_n \cos(nx) + b_n \sin(nx))$$

S_n is the element of $\text{Span}(\mathcal{F}_n)$ which minimizes $\|f - S_n\|^2$.

S_n is the function of the form :

$$\frac{a_0}{2} + (a_1 \cos(x) + b_1 \sin(x)) + \dots + (a_n \cos(x) + b_n \sin(x))$$

The coefficient $(a_0, \dots, a_n, b_n)$ which leads to the smallest $\|f - S_n\|^2$ are :

$$\begin{cases} a_n = \langle f, \cos(nx) \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(x) dx \\ b_n = \langle f, \sin(nx) \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \end{cases}$$

Where S_n is the approximation of f and Error = $f - S_n$.

We get the smallest Error function with :

$$\forall n \in \mathbb{N}, \begin{cases} a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \\ b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \end{cases}$$

6.2 Theorems

6.2.1 Parseval Theorems

If f is piecewise of class $\mathcal{C}^1$, then $(S_n) \xrightarrow{L^2} f$ that is, $\|S_n - f\| \xrightarrow{n \rightarrow +\infty} 0$.

Where :

$$\|S_n - f\| = \sqrt{\frac{1}{\pi} \int_{-\pi}^{\pi} (S_n(x) - f(x))^2 dx}$$

Consequence :

$$f = S_n + \text{Error} \implies \|f\|^2 = \|S_n\|^2 + \|\text{Error}\|^2$$

Here, $\|\text{Error}\|^2 \xrightarrow{n \rightarrow +\infty} 0$ and $\|f\|^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx$.

Also, $\|S_n\|^2 = \frac{a_0^2}{2} + (a_1^2 + b_1^2) + \dots + (a_n^2 + b_n^2)$

Thus,

$$\frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx = \frac{a_0^2}{2} + \sum_{n=1}^{+\infty} (a_n^2 + b_n^2)$$

6.3 Dirichlet theorem

$$FS(f) = (S_n) \xrightarrow{\text{P.W}} \tilde{f}$$

Where $\tilde{f}$ is the regularized f.

$$\forall x \in \mathbb{R}, \tilde{f}(x) = \begin{cases} x & \text{if } f \text{ continuous at } x \\ \frac{f(x^-) + f(x^+)}{2} & \text{if } f \text{ not continuous at } x \end{cases}$$

$$f = S_n + \text{Error}$$

7 Complex Notation

$$c_n = \frac{a_n - ib_n}{2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

Check :

$$\begin{aligned} & \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) (\cos(nx) - i \sin(nx)) dx \\ &= \frac{1}{\pi} \int f(x) \frac{\cos(nx)}{2} - f(x) \frac{\sin(nx)}{2} dx = \frac{a_n}{2} - \frac{b_n}{2} \end{aligned}$$

Thus :

$$FS(f) = \sum_{n=-\infty}^{+\infty} c_n e^{inx}$$

We also have :

$$c_0 e^{-i \times 0 \times x} = c_0 = \frac{a_0}{2} + \frac{b_0}{2}$$

$$\begin{aligned} & \forall n > 0, c_{-n} e^{-inx} + c_n e^{inx} \\ &= \left(\frac{a_n}{2} + \frac{ib_n}{2}\right)(\cos(-nx) + i\sin(-nx)) + \left(\frac{a_n}{2} - \frac{ib_n}{2}\right)(\cos(nx) + i\sin(nx)) \end{aligned}$$

Note that :

$$\begin{aligned} C_{-n} &= \overline{C_n} = \frac{a_n}{2} - i\frac{b_n}{2} \\ &= \frac{a_n}{2}\cos(nx) - i\frac{a_n}{2}\sin(nx) + i\frac{b_n}{2}\cos(nx) + \frac{b_n}{2}\sin(nx) \\ &\quad + \frac{a_n}{2}\cos(nx) + i\frac{a_n}{2}\sin(nx) - i\frac{b_n}{2}\cos(nx) + \frac{b_n}{2}\sin(nx) \\ &= a_n\cos(nx) + b_n\sin(nx) \end{aligned}$$